Noise as structure

Introduction

Noise is not a residual term to be minimized, but an integral part of the model itself.

When I was a child, I wanted to understand physics. Even before learning mathematics properly, I had the intuition that everything could be described by equations or numbers at least. At first, one learns simple equations: linear laws, ordinary differential equations, clean deterministic systems.

But as soon as we try to understand physics more deeply, not just to model simplified or idealized systems, we inevitably enter a darker territory: stochastic processes and randomness.

Uncertainty does not only appear in physics. It is central to biology (exploration, robustness, variability), to artificial intelligence (diffusion models, Bayesian reinforcement learning), and to neuroscience, where neuronal variability is not a bug but a defining feature of neural computation.

In textbooks, stochasticity often appears as a technical afterthought: just add a Wiener term to the equation and you are done. The random term is usually presented as a nuisance, something intimidating but ultimately secondary.

Yet this stochastic term often structures the entire model. It shapes the dynamics, the stationary distributions, the stability properties, and even what can be learned or inferred from data. In many systems, removing noise does not reveal the "true" behavior, it destroys it.

Deterministic models: when ODEs are enough

Let us start with the simplest setting: deterministic ordinary differential equations. Consider a basic RC electrical circuit. If $V(t)$ is the voltage across the capacitor, the dynamics are governed by

$$RC \cdot \frac{dV(t)}{dt} + V(t) = V_{in}(t).$$

This model works remarkably well under strong assumptions:

- the components are ideal,
- the environment is perfectly controlled,
- the initial condition V(0) is known exactly.

Under these conditions, the system is fully deterministic: given an initial condition, the trajectory is uniquely determined. For many engineering problems, this is more than sufficient.

But this apparent success hides a deeper issue: real systems are never perfectly isolated, and their initial conditions are never known exactly.

Adding noise: stochastic differential equations

As soon as uncertainty becomes intrinsic to the system, deterministic ODEs are no longer adequate. A canonical example comes from quantitative finance. In option pricing, asset prices are not just unknown — they are fundamentally random. The standard model for a stock price S_t is not an ODE but a stochastic differential equation (SDE):

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

, where μ is the drift (expected growth), σ is the volatility and W_t is a Wiener process (Brownian motion).

Here, the noise term does not represent measurement error. It is the market. This equation already reveals a crucial idea: the drift encodes structure while the diffusion encodes uncertainty, both are equally fundamental.

At this point, a subtle but important distinction appears: Itô vs Stratonovich calculus. While equivalent in many physical limits, they correspond to different assumptions about how noise interacts with the system, and they can lead to different effective dynamics.

From trajectories to distributions: the Fokker–Planck equation

When dealing with stochastic systems, a single trajectory is rarely meaningful. Instead of asking "where is the particle?", we ask: What is the probability that the particle is at position x at time t ? For an SDE of the form

$$dx_t = f(x_t)dt + g(x_t)dW_t$$

, the evolution of the probability density

$$p(x, t)$$

is governed by the Fokker–Planck equation:

$$\frac{\partial p(x,t)}{\partial t} = -\nabla \cdot (f(x)p(x,t)) + \frac{1}{2}\nabla^2(g^2(x)p(x,t))$$

This equation shifts the perspective: dynamics are no longer about trajectories, they are about probability flows. Stationary distributions, metastability, and long-term behavior are often invisible at the trajectory level but become clear at the *distribution* level.

Counter-intuitive effects: when noise helps

Once noise is taken seriously, surprising phenomena emerge. One example is escape from local minima. In a deterministic system, a particle trapped in a potential well remains there forever. With noise, escape becomes possible, with rates described by Kramers' theory. Another striking phenomenon is stochastic resonance, where adding noise enhances the response of a system to weak signals. In such cases, noise does not obscure structure — it reveals it. These effects show that noise is not merely tolerated by systems, it can be actively exploited.

Learning under uncertainty: Bayesian filtering

In learning and inference, uncertainty often comes from hidden states. A general state-space model is written as:

$$x_{t+1} = f(x_t) + \varepsilon_t$$

$$y_t = h(x_t) + \eta_t$$

, where x_t is latent and only indirect observations y_t are available.

Bayesian filtering aims to estimate the belief distribution:

$$p(x_t \mid y_{1:t})$$

Kalman filters solve this exactly for linear-Gaussian systems. Particle filters approximate it for nonlinear or non-Gaussian cases. Here, a crucial distinction appears: aleatoric uncertainty: irreducible randomness, epistemic uncertainty: uncertainty due to lack of knowledge.

Learning consists precisely in transforming epistemic uncertainty into structure.

Modern diffusion models: learning by reversing noise

Today, diffusion models generate images, audio, and text by learning how to invert a diffusion process. A simple forward process adds noise: $dx_t = \sqrt{\beta(t)}dW_t$, gradually destroying structure.

The model learns the reverse dynamics:

$$dx_t = \left[f(x_t, t) - \beta(t)\nabla \log p_{t(x_t)}\right]dt + \beta(t)dW_t.$$

Generation is thus a controlled descent through noise, structure emerging from randomness. This is not a metaphor: it is literally stochastic dynamics run backward in time.

Conclusion: noise is the model

Across physics, finance, biology, and neuroscience, the same lesson appears:

Noise is not an error term to be removed. It defines what is stable, what is learnable, and what is possible.

In many systems, removing noise does not reveal the truth, it removes the phenomenon itself.